

PLATE BOUNDARIES

When plates meet, the collisions can form new ocean trenches or mountain ranges—or cause huge volcanic eruptions or earthquakes.

-  Divergent or transform plate boundary
-  Convergent plate boundary
-  Convergent plate boundary creating a deep-sea trench

Caribbean/North American boundary

This transform border is a region of frequent earthquakes, tsunamis, and volcanic eruptions.

African Plate

East African Rift

This series of great valleys fringed in places by volcanoes is where the African Plate is in the process of splitting into two new tectonic plates.

Mid-Atlantic Ridge

The African Plate is moving northeast at about 0.85 in (2.15 cm) per year, creating the Mid-Atlantic Ridge—a chain of volcanic mountains.

South American Plate

Scotia Plate

Eurasian Plate

Indian Plate

Arabian Plate

Himalayas

The Himalayan mountain range was formed when two plates of continental crust collided. The land masses crumpled and formed enormous, jagged mountain peaks.

Types of boundaries

The huge slabs of rock that cover Earth are called tectonic plates. Where the plates meet, they form different kinds of boundaries, depending on the type of crust they are made of and the directions in which they move.

Convergent boundary

Where two plates collide. In some cases, one plate is pushed under the other; this is called subduction.

Divergent boundary

Where plates move apart, molten magma rises from the mantle to fill the gap, building a mid-ocean ridge.

Transform boundary

Transform boundaries are formed where the two plates scrape past each other in a sideways motion.